

Representation Theory

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Chapter 1

Representations and Characters

Throughout this chapter, G denotes a finite group and all vector spaces are finite-dimensional over $\mathbb{C}$.

1.1 Linear Representations and Invariant Subspaces

Definition 1.1. A *linear representation* of G on a vector space V is a group homomorphism

$$\rho: G \longrightarrow \mathrm{GL}(V).$$

When ρ is understood, we also say that V is a representation space of G .

Definition 1.2. Let

$$\rho: G \rightarrow \mathrm{GL}(V), \quad \rho': G \rightarrow \mathrm{GL}(V')$$

be two representations. They are said to be *equivalent*, or *isomorphic*, if there exists a linear isomorphism $T: V \rightarrow V'$ such that

$$T\rho(g) = \rho'(g)T$$

for every $g \in G$. Such a map is called an *intertwining operator*.

Definition 1.3. Let $\rho: G \rightarrow \mathrm{GL}(V)$ be a representation. A vector subspace $W \subseteq V$ is called *G -invariant* if

$$\rho(g)W \subseteq W$$

for every $g \in G$. Since each $\rho(g)$ is invertible, this is equivalent to $\rho(g)W = W$. The restricted maps

$$\rho|_W(g) = \rho(g)|_W$$

define a representation of G on W , called a *subrepresentation* of V .

Theorem 1.4 (Maschke's theorem). *Let $W \subseteq V$ be a G -invariant subspace. Then W admits a G -invariant complement: there exists a G -invariant subspace W' such that*

$$V = W \oplus W'.$$

Proof. Choose an arbitrary vector-space complement U of W and let $p: V \rightarrow W$ be the projection with kernel U . Define the averaged map

$$p^0 = \frac{1}{|G|} \sum_{g \in G} \rho(g)p\rho(g)^{-1}.$$

Since W is G -invariant, every summand maps V into W , hence so does p^0 . Moreover, if $w \in W$, then

$$\rho(g)p\rho(g)^{-1}w = w$$

for every $g \in G$, and therefore $p^0|_W = \text{id}_W$. It follows that $(p^0)^2 = p^0$ and

$$V = W \oplus \ker(p^0).$$

For $s \in G$ we have

$$\begin{aligned} \rho(s)p^0\rho(s)^{-1} &= \frac{1}{|G|} \sum_{g \in G} \rho(sg)p\rho(sg)^{-1} \\ &= p^0, \end{aligned}$$

so p^0 commutes with the action of G . Hence $\ker(p^0)$ is G -invariant, and it is the required complement. $\square$

Definition 1.5. A nonzero representation V is called *irreducible* if its only G -invariant subspaces are 0 and V . Equivalently, V cannot be written as a direct sum of two nonzero subrepresentations.

Theorem 1.6 (Complete reducibility). *Every representation of G is a direct sum of irreducible representations.*

Proof. We argue by induction on $\dim V$. There is nothing to prove when $V = 0$ or when V is irreducible. Otherwise, choose a nonzero proper invariant subspace $W \subsetneq V$. By Maschke's theorem,

$$V = W \oplus W'$$

for some invariant complement W' . Both W and W' have smaller dimension than V , so the induction hypothesis decomposes each of them into irreducible representations. Combining the two decompositions gives the result. $\square$

1.2 Constructions of Representations

1.2.1 Direct sums

Let $\rho_1: G \rightarrow \text{GL}(V_1)$ and $\rho_2: G \rightarrow \text{GL}(V_2)$ be representations. Their *direct sum* is the representation

$$\rho_1 \oplus \rho_2: G \longrightarrow \text{GL}(V_1 \oplus V_2)$$

defined by

$$(\rho_1 \oplus \rho_2)(g)(v_1, v_2) = (\rho_1(g)v_1, \rho_2(g)v_2).$$

With respect to bases of V_1 and V_2 , its matrix is block diagonal:

$$(\rho_1 \oplus \rho_2)(g) = \begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix}.$$

1.2.2 Tensor products

The *tensor product representation* on $V_1 \otimes V_2$ is defined by

$$(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) = \rho_1(g)v_1 \otimes \rho_2(g)v_2.$$

This indeed gives a representation because

$$(\rho_1 \otimes \rho_2)(gh) = (\rho_1 \otimes \rho_2)(g)(\rho_1 \otimes \rho_2)(h).$$

Let (e_i) be a basis of V_1 and (f_j) a basis of V_2 , and write

$$\rho_1(g)e_j = \sum_i r_{ij}^{(1)}(g)e_i, \quad \rho_2(g)f_\ell = \sum_k r_{k\ell}^{(2)}(g)f_k.$$

Then

$$(\rho_1 \otimes \rho_2)(g)(e_j \otimes f_\ell) = \sum_{i,k} r_{ij}^{(1)}(g)r_{k\ell}^{(2)}(g)e_i \otimes f_k.$$

Thus the matrix of the tensor product representation is the Kronecker product of the matrices of $\rho_1(g)$ and $\rho_2(g)$.

1.2.3 Symmetric and alternating squares

Suppose now that $V_1 = V_2 = V$. Define

$$\tau: V \otimes V \longrightarrow V \otimes V, \quad \tau(v \otimes w) = w \otimes v.$$

Since $\tau^2 = \text{id}$, its eigenvalues are 1 and -1 . We set

$$\begin{aligned} \text{Sym}^2(V) &= \{z \in V \otimes V \mid \tau(z) = z\}, \\ \bigwedge^2 V &= \{z \in V \otimes V \mid \tau(z) = -z\}. \end{aligned}$$

Because the ground field has characteristic different from 2,

$$V \otimes V = \text{Sym}^2(V) \oplus \bigwedge^2 V.$$

The flip map τ commutes with $\rho(g) \otimes \rho(g)$, so both summands are G -invariant.

If $(e_1, \dots, e_n)$ is a basis of V , then a basis of $\text{Sym}^2(V)$ is

$$e_i \otimes e_i \quad (1 \leq i \leq n), \quad e_i \otimes e_j + e_j \otimes e_i \quad (1 \leq i < j \leq n),$$

and a basis of $\bigwedge^2 V$ is

$$e_i \otimes e_j - e_j \otimes e_i \quad (1 \leq i < j \leq n).$$

In particular,

$$\dim \text{Sym}^2(V) = \frac{n(n+1)}{2}, \quad \dim \bigwedge^2 V = \frac{n(n-1)}{2}.$$

1.2.4 Dual representations

Let $V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$. The *dual representation* ρ^* is defined by

$$\rho^*(g) = (\rho(g)^{-1})^*.$$

Equivalently,

$$(\rho^*(g)\lambda)(v) = \lambda(\rho(g)^{-1}v)$$

for $\lambda \in V^*$ and $v \in V$. It is characterized by the identity

$$\langle \rho(g)v, \rho^*(g)\lambda \rangle = \langle v, \lambda \rangle.$$

The homomorphism property follows from

$$\rho^*(gh) = (\rho(h)^{-1}\rho(g)^{-1})^* = \rho^*(g)\rho^*(h).$$

1.3 Characters of Representations

Definition 1.7. Let $A: V \rightarrow V$ be linear. If (a_{ij}) is its matrix in a basis of V , its *trace* is

$$\text{Tr}(A) = \sum_i a_{ii}.$$

The trace is independent of the chosen basis, since

$$\text{Tr}(PAP^{-1}) = \text{Tr}(A)$$

for every invertible matrix P .

Definition 1.8. Let $\rho: G \rightarrow \text{GL}(V)$ be a representation. Its *character* is the function

$$\chi_\rho: G \longrightarrow \mathbb{C}, \quad \chi_\rho(g) = \text{Tr}(\rho(g)).$$

The integer $\dim V$ is also called the *degree* of the character.

Proposition 1.9 (Basic properties of characters). *Let χ be the character of a representation V of degree n . Then:*

1. $\chi(1) = n$;
2. $\chi(g^{-1}) = \overline{\chi(g)}$ for every $g \in G$;
3. $\chi(hgh^{-1}) = \chi(g)$ for every $g, h \in G$.

Proof. The first assertion is immediate. For the second, let m be the order of g . Then $\rho(g)^m = \text{id}$, so the minimal polynomial of $\rho(g)$ divides $x^m - 1$. Since $x^m - 1$ has distinct roots over $\mathbb{C}$, the operator $\rho(g)$ is diagonalizable. If its eigenvalues are $\lambda_1, \dots, \lambda_n$, then $\lambda_i^m = 1$, and hence $\lambda_i^{-1} = \overline{\lambda_i}$. Therefore

$$\chi(g^{-1}) = \sum_i \lambda_i^{-1} = \sum_i \overline{\lambda_i} = \overline{\chi(g)}.$$

Finally,

$$\begin{aligned} \chi(hgh^{-1}) &= \text{Tr}(\rho(h)\rho(g)\rho(h)^{-1}) \\ &= \text{Tr}(\rho(g)) = \chi(g). \end{aligned}$$

□

Proposition 1.10. *Let V_1, V_2 have characters χ_1, χ_2 . Then*

$$\chi_{V_1 \oplus V_2} = \chi_1 + \chi_2, \quad \chi_{V_1 \otimes V_2} = \chi_1 \chi_2.$$

Proof. The first formula follows from the trace of a block diagonal matrix. For the second, use the notation of the tensor-product construction. The diagonal entries of the matrix of $\rho_1(g) \otimes \rho_2(g)$ are $r_{ii}^{(1)}(g)r_{jj}^{(2)}(g)$, and hence

$$\begin{aligned} \chi_{V_1 \otimes V_2}(g) &= \sum_{i,j} r_{ii}^{(1)}(g)r_{jj}^{(2)}(g) \\ &= \left(\sum_i r_{ii}^{(1)}(g) \right) \left(\sum_j r_{jj}^{(2)}(g) \right) \\ &= \chi_1(g)\chi_2(g). \end{aligned}$$

□

Proposition 1.11 (Characters of the symmetric and alternating squares). *Let V have character χ . Then*

$$\chi_{\text{Sym}^2(V)}(g) = \frac{1}{2}(\chi(g)^2 + \chi(g^2))$$

and

$$\chi_{\Lambda^2 V}(g) = \frac{1}{2}(\chi(g)^2 - \chi(g^2)).$$

In particular,

$$\chi_{\text{Sym}^2(V)} + \chi_{\Lambda^2 V} = \chi^2.$$

Proof. Fix $g \in G$. Since $\rho(g)$ has finite order, it is diagonalizable. Choose an eigenbasis $e_1, \dots, e_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$. On the basis vectors of $\text{Sym}^2(V)$, the eigenvalues are

$$\lambda_i^2 \quad \text{and} \quad \lambda_i \lambda_j \quad (i < j),$$

whereas on $\Lambda^2 V$ they are $\lambda_i \lambda_j$ for $i < j$. Therefore

$$\begin{aligned} \chi_{\text{Sym}^2(V)}(g) &= \sum_i \lambda_i^2 + \sum_{i < j} \lambda_i \lambda_j \\ &= \frac{1}{2} \left(\left(\sum_i \lambda_i \right)^2 + \sum_i \lambda_i^2 \right) \\ &= \frac{1}{2}(\chi(g)^2 + \chi(g^2)), \end{aligned}$$

and similarly

$$\begin{aligned} \chi_{\Lambda^2 V}(g) &= \sum_{i < j} \lambda_i \lambda_j \\ &= \frac{1}{2} \left(\left(\sum_i \lambda_i \right)^2 - \sum_i \lambda_i^2 \right) \\ &= \frac{1}{2}(\chi(g)^2 - \chi(g^2)). \end{aligned}$$

□

Corollary 1.12. *Let V and W have characters χ and ψ . Then*

$$\begin{aligned}\chi_{\text{Sym}^2(V \oplus W)} &= \chi_{\text{Sym}^2 V} + \chi\psi + \chi_{\text{Sym}^2 W}, \\ \chi_{\wedge^2(V \oplus W)} &= \chi_{\wedge^2 V} + \chi\psi + \chi_{\wedge^2 W}.\end{aligned}$$

Proof. Use the G -equivariant decompositions

$$\begin{aligned}\text{Sym}^2(V \oplus W) &\cong \text{Sym}^2 V \oplus (V \otimes W) \oplus \text{Sym}^2 W, \\ \bigwedge^2(V \oplus W) &\cong \bigwedge^2 V \oplus (V \otimes W) \oplus \bigwedge^2 W,\end{aligned}$$

and apply Proposition 1.10. □

Proposition 1.13. *If V has character χ , then its dual representation has character*

$$\chi_{V^*}(g) = \chi(g^{-1}) = \overline{\chi(g)}.$$

Proof. The eigenvalues of $\rho^*(g)$ are the inverses of the eigenvalues of $\rho(g)$. Hence

$$\chi_{V^*}(g) = \chi(g^{-1}),$$

and Proposition 1.9 gives the second equality. □

1.4 Schur's Lemma and Averaging Operators

Proposition 1.14 (Schur's lemma). *Let V_1, V_2 be irreducible representations and let $T: V_1 \rightarrow V_2$ satisfy*

$$T\rho_1(g) = \rho_2(g)T$$

for all $g \in G$. Then:

1. *if $V_1 \not\cong V_2$, then $T = 0$;*
2. *if $V_1 = V_2$ and $\rho_1 = \rho_2$, then $T = \lambda \text{id}_{V_1}$ for some $\lambda \in \mathbb{C}$.*

Proof. The kernel of T is a G -invariant subspace of V_1 , and the image of T is a G -invariant subspace of V_2 . If $T \neq 0$, irreducibility gives

$$\ker T = 0, \quad \text{im } T = V_2,$$

so T is an isomorphism. This proves the first assertion.

For the second, choose an eigenvalue λ of T , which exists because the field is $\mathbb{C}$. The map $T - \lambda \text{id}_{V_1}$ is again an intertwining operator and has nonzero kernel. By irreducibility its kernel is all of V_1 , so $T = \lambda \text{id}_{V_1}$. □

Definition 1.15. For a linear map $h: V_1 \rightarrow V_2$, define its *average* by

$$h^0 = \frac{1}{|G|} \sum_{g \in G} \rho_2(g)^{-1} h \rho_1(g).$$

Proposition 1.16. *The averaged map h^0 is an intertwining operator. Consequently, if V_1, V_2 are irreducible, then:*

1. if $V_1 \not\cong V_2$, then $h^0 = 0$;
2. if $V_1 = V_2 = V$ and $\rho_1 = \rho_2$, then

$$h^0 = \frac{\text{Tr}(h)}{\dim V} \text{id}_V.$$

Proof. For $s \in G$,

$$\begin{aligned} \rho_2(s)^{-1} h^0 \rho_1(s) &= \frac{1}{|G|} \sum_{g \in G} \rho_2(gs)^{-1} h \rho_1(gs) \\ &= h^0. \end{aligned}$$

Thus $h^0 \rho_1(s) = \rho_2(s) h^0$, so h^0 is an intertwiner. The first conclusion follows from Schur's lemma. In the second case, Schur's lemma gives $h^0 = \lambda \text{id}_V$. Since trace is invariant under conjugation,

$$\text{Tr}(h^0) = \frac{1}{|G|} \sum_{g \in G} \text{Tr}(h) = \text{Tr}(h).$$

Hence $\lambda \dim V = \text{Tr}(h)$. □

Lemma 1.17 (Averaging onto invariant vectors). *Let E be a representation of G with character χ_E . Then*

$$\dim E^G = \frac{1}{|G|} \sum_{g \in G} \chi_E(g),$$

where

$$E^G = \{v \in E \mid \rho(g)v = v \text{ for every } g \in G\}.$$

Proof. Set

$$P = \frac{1}{|G|} \sum_{g \in G} \rho(g).$$

For every $s \in G$ we have $\rho(s)P = P$, so $\text{im } P \subseteq E^G$. Conversely, P acts as the identity on E^G , and therefore P is the projection of E onto E^G . Thus

$$\dim E^G = \text{Tr}(P) = \frac{1}{|G|} \sum_{g \in G} \chi_E(g).$$

□

1.5 Orthogonality Relations

For complex-valued functions φ, ψ on G , define

$$\langle \varphi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \varphi(g) \overline{\psi(g)}.$$

This is the standard Hermitian inner product on the space of functions $G \rightarrow \mathbb{C}$.

1.5.1 Orthogonality of matrix coefficients

Let V_1, V_2 be irreducible representations. Choose bases and write

$$\rho_\alpha(g)e_j^{(\alpha)} = \sum_i r_{ij}^{(\alpha)}(g)e_i^{(\alpha)} \quad (\alpha = 1, 2).$$

Theorem 1.18 (Orthogonality of matrix coefficients). *If $V_1 \not\cong V_2$, then*

$$\frac{1}{|G|} \sum_{g \in G} r_{i_2 j_2}^{(2)}(g^{-1}) r_{j_1 i_1}^{(1)}(g) = 0$$

for all indices. If $V_1 = V_2 = V$ and the same basis is used on both sides, then, with $n = \dim V$,

$$\frac{1}{|G|} \sum_{g \in G} r_{i_2 j_2}(g^{-1}) r_{j_1 i_1}(g) = \frac{1}{n} \delta_{i_1 i_2} \delta_{j_1 j_2}.$$

Proof. Let $h: V_1 \rightarrow V_2$ have matrix $(x_{j_2 j_1})$. By direct matrix multiplication, the (i_2, i_1) -entry of its average is

$$(h^0)_{i_2 i_1} = \frac{1}{|G|} \sum_{g \in G} \sum_{j_1, j_2} r_{i_2 j_2}^{(2)}(g^{-1}) x_{j_2 j_1} r_{j_1 i_1}^{(1)}(g).$$

If $V_1 \not\cong V_2$, Proposition 1.16 gives $h^0 = 0$ for every choice of the matrix $(x_{j_2 j_1})$, so every coefficient of every $x_{j_2 j_1}$ is zero. This proves the first formula.

In the second case,

$$h^0 = \frac{\text{Tr}(h)}{n} \text{id}_V.$$

The coefficient of $x_{j_2 j_1}$ in the (i_2, i_1) -entry of the right-hand side is

$$\frac{1}{n} \delta_{j_1 j_2} \delta_{i_1 i_2},$$

which gives the stated identity. □

1.5.2 Orthogonality of irreducible characters

Theorem 1.19 (Character orthogonality). *Let χ and ψ be irreducible characters. Then*

$$\langle \chi, \psi \rangle = \begin{cases} 1, & \chi = \psi, \\ 0, & \chi \neq \psi. \end{cases}$$

Thus the irreducible characters form an orthonormal set.

Proof. Using $\overline{\psi(g)} = \psi(g^{-1})$ and expanding both traces in terms of matrix coefficients, we obtain

$$\begin{aligned} \langle \chi, \psi \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi(g) \psi(g^{-1}) \\ &= \sum_{i, j} \frac{1}{|G|} \sum_{g \in G} r_{ii}^{(1)}(g) r_{jj}^{(2)}(g^{-1}). \end{aligned}$$

If the representations are nonisomorphic, every summand is zero by Theorem 1.18. If they are the same irreducible representation of dimension n , the same theorem gives

$$\langle \chi, \chi \rangle = \sum_{i,j} \frac{1}{n} \delta_{ij} = 1.$$

□

Remark 1.20. There is also a conceptual proof. The space $\text{Hom}_{\mathbb{C}}(V, W)$ is a representation under

$$(g \cdot T) = \rho_W(g) T \rho_V(g)^{-1},$$

and its character is $\chi_W \overline{\chi_V}$. By Lemma 1.17,

$$\langle \chi_W, \chi_V \rangle = \dim \text{Hom}_G(V, W).$$

Schur's lemma then gives the orthogonality relations.

Theorem 1.21 (Multiplicity formula). *Suppose*

$$V = U_1 \oplus \cdots \oplus U_m$$

is a decomposition into irreducible representations, and let φ be the character of V . If W is irreducible with character χ , then the number of summands U_j isomorphic to W is

$$\langle \varphi, \chi \rangle.$$

Proof. Write χ_j for the character of U_j . Then

$$\varphi = \chi_1 + \cdots + \chi_m.$$

By the orthogonality theorem,

$$\langle \varphi, \chi \rangle = \sum_{j=1}^m \langle \chi_j, \chi \rangle,$$

and each summand is 1 precisely when $U_j \cong W$, and 0 otherwise. □

Corollary 1.22. *Two representations with the same character are isomorphic.*

Proof. For every irreducible representation W , the multiplicity formula shows that its multiplicity is determined by the character. Hence the two representations have the same irreducible summands with the same multiplicities, and complete reducibility gives an isomorphism. □

Theorem 1.23 (Irreducibility criterion). *Let φ be the character of a representation V . Then*

$$\langle \varphi, \varphi \rangle \in \mathbb{Z}_{\geq 0},$$

and

$$\langle \varphi, \varphi \rangle = 1$$

if and only if V is irreducible.

Proof. Let $\chi_1, \dots, \chi_r$ be the distinct irreducible characters occurring in V , with multiplicities $m_1, \dots, m_r$. Then

$$\varphi = \sum_{i=1}^r m_i \chi_i.$$

By orthogonality,

$$\langle \varphi, \varphi \rangle = \sum_{i=1}^r m_i^2.$$

This is a nonnegative integer, and it equals 1 exactly when one multiplicity is 1 and all the others are zero, which is equivalent to irreducibility. $\square$

1.6 Canonical Decomposition into Isotypic Components

Let $W_1, \dots, W_r$ be pairwise nonisomorphic irreducible representations of G , with characters $\chi_1, \dots, \chi_r$ and degrees $n_i = \dim W_i$. Suppose that a representation V has an irreducible decomposition

$$V = U_1 \oplus \dots \oplus U_m.$$

For each i , let V_i be the direct sum of those U_j which are isomorphic to W_i . Then

$$V = V_1 \oplus \dots \oplus V_r.$$

At first sight the subspaces V_i appear to depend on the chosen irreducible decomposition. The following theorem shows that they do not.

Theorem 1.24 (Canonical isotypic decomposition). *For each i , define*

$$p_i = \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g) = \frac{n_i}{|G|} \sum_{g \in G} \chi_i(g^{-1}) \rho(g).$$

Then p_i is the G -equivariant projection of V onto V_i . In particular,

$$p_i^2 = p_i, \quad p_i p_j = 0 \quad (i \neq j), \quad \sum_{i=1}^r p_i = \text{id}_V,$$

and the decomposition

$$V = V_1 \oplus \dots \oplus V_r$$

is independent of the chosen decomposition of V into irreducible summands.

Proof. Since χ_i is constant on conjugacy classes, for every $h \in G$ we have

$$\begin{aligned} \rho(h) p_i \rho(h)^{-1} &= \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(h g h^{-1}) \\ &= p_i. \end{aligned}$$

Thus p_i is G -equivariant.

Restrict p_i to an irreducible subrepresentation $U \subseteq V$ with character χ_j . By Schur's lemma,

$$p_i|_U = \lambda \text{id}_U$$

for some $\lambda \in \mathbb{C}$. Taking traces gives

$$\begin{aligned}\lambda n_j &= \text{Tr}(p_i|_U) \\ &= \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \chi_j(g) \\ &= n_i \langle \chi_j, \chi_i \rangle.\end{aligned}$$

By character orthogonality, this is 0 if $i \neq j$, and it is $n_i = n_j$ if $i = j$. Hence

$$p_i|_U = \begin{cases} \text{id}_U, & U \cong W_i, \\ 0, & U \not\cong W_i. \end{cases}$$

Therefore p_i is exactly the projection onto the sum of all irreducible summands isomorphic to W_i . The identities for the p_i follow by checking them on every irreducible summand of V . Since each p_i is defined directly from ρ and χ_i , its image is independent of any chosen irreducible decomposition. $\square$

Definition 1.25. The subspace

$$V_i = \text{im}(p_i)$$

is called the W_i -isotypic component of V . Equivalently, it is the sum of all subrepresentations of V isomorphic to W_i .

Remark 1.26. The decomposition into individual irreducible subspaces need not be unique. For example, if

$$V \cong W \oplus W,$$

there may be many different subspaces of V isomorphic to W which can be used as summands. What is canonical is their total isotypic component, which in this example is all of V .

1.6.1 Multiplicity spaces

Proposition 1.27. *Let*

$$M_i = \text{Hom}_G(W_i, V).$$

Then

$$\dim M_i = \langle \chi_V, \chi_i \rangle,$$

and the evaluation map

$$\begin{aligned}\text{ev}_i: W_i \otimes M_i &\longrightarrow V_i, \\ w \otimes f &\longmapsto f(w)\end{aligned}$$

is a G -equivariant isomorphism. Consequently,

$$V \cong \bigoplus_{i=1}^r W_i \otimes \text{Hom}_G(W_i, V).$$

Proof. By Schur's lemma and complete reducibility, $\dim \text{Hom}_G(W_i, V)$ is exactly the number of copies of W_i in an irreducible decomposition of V . By the multiplicity formula this number is $\langle \chi_V, \chi_i \rangle$.

The evaluation map is G -equivariant because, for $g \in G$ and $f \in \text{Hom}_G(W_i, V)$,

$$\text{ev}_i(\rho_i(g)w \otimes f) = f(\rho_i(g)w) = \rho(g)f(w).$$

Its image is contained in V_i . Conversely, every irreducible copy of W_i inside V is the image of an intertwining map $W_i \rightarrow V$, so the image of ev_i contains every such copy and hence equals V_i . Finally,

$$\dim(W_i \otimes M_i) = n_i \dim M_i = \dim V_i,$$

so the surjective evaluation map is an isomorphism. $\square$

1.7 Permutation and Regular Representations

1.7.1 Permutation representations

Definition 1.28. Suppose that G acts on a finite set X . Let V_X be the vector space with basis $(e_x)_{x \in X}$. The rule

$$\rho(g)e_x = e_{gx}$$

defines a representation of G , called the *permutation representation* associated with X .

Proposition 1.29. *The character χ_X of the permutation representation satisfies*

$$\chi_X(g) = |X^g|,$$

where

$$X^g = \{x \in X \mid gx = x\}.$$

Proof. The matrix of $\rho(g)$ in the basis $(e_x)_{x \in X}$ is a permutation matrix. Its diagonal entry corresponding to x is 1 exactly when $gx = x$, and is 0 otherwise. Hence its trace is the number of fixed points of g . $\square$

Proposition 1.30 (Orbit-counting formula). *The multiplicity of the trivial representation in V_X is the number of G -orbits on X . Equivalently,*

$$\#(G \backslash X) = \frac{1}{|G|} \sum_{g \in G} |X^g| = \langle \chi_X, \mathbf{1}_G \rangle = \dim V_X^G.$$

Proof. An invariant vector

$$v = \sum_{x \in X} a_x e_x$$

has coefficients which are constant on every orbit. Therefore the orbit sums

$$\sum_{x \in \mathcal{O}} e_x, \quad \mathcal{O} \in G \backslash X,$$

form a basis of V_X^G , so $\dim V_X^G = \#(G \backslash X)$. Lemma 1.17 and Proposition 1.29 give the remaining equalities. $\square$

If the action is transitive, the trivial representation occurs exactly once. In this case

$$V_X = \mathbb{C} \left(\sum_{x \in X} e_x \right) \oplus V_0,$$

where

$$V_0 = \left\{ \sum_{x \in X} a_x e_x \mid \sum_{x \in X} a_x = 0 \right\}.$$

The representation V_0 contains no copy of the trivial representation.

Definition 1.31. Assume that G acts transitively on X and $|X| \geq 2$. The action is called *doubly transitive* if for any $x \neq y$ and $x' \neq y'$ in X , there exists $g \in G$ such that

$$gx = x', \quad gy = y'.$$

Proposition 1.32. Let G act transitively on X , let χ_X be the permutation character, and write

$$\chi_X = \mathbf{1}_G + \theta,$$

where θ is the character of V_0 . The following are equivalent:

1. the action of G on X is doubly transitive;
2. the diagonal action of G on $X \times X$ has exactly two orbits;
3. $\langle \chi_X, \chi_X \rangle = 2$;
4. V_0 is irreducible.

Proof. The diagonal

$$\Delta = \{(x, x) \mid x \in X\}$$

is one orbit because the action on X is transitive. The action is doubly transitive precisely when all pairs (x, y) with $x \neq y$ form one further orbit. Thus (1) and (2) are equivalent.

The character of the permutation representation on $X \times X$ is χ_X^2 , because

$$|(X \times X)^g| = |X^g|^2.$$

By Proposition 1.30, the number of orbits on $X \times X$ is

$$\langle \chi_X^2, \mathbf{1}_G \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_X(g)^2 = \langle \chi_X, \chi_X \rangle,$$

since χ_X is real-valued. Hence (2) and (3) are equivalent.

Finally, $\langle \mathbf{1}_G, \theta \rangle = 0$, so

$$\langle \chi_X, \chi_X \rangle = 1 + \langle \theta, \theta \rangle.$$

By the irreducibility criterion, this equals 2 precisely when V_0 is irreducible. Thus (3) and (4) are equivalent. $\square$

1.7.2 The regular representation

Definition 1.33. Let V have basis $(e_t)_{t \in G}$. For $g \in G$, define

$$\rho_{\text{reg}}(g)e_t = e_{gt}.$$

This is the *left regular representation* of G .

Its character χ_{reg} is

$$\chi_{\text{reg}}(g) = \begin{cases} |G|, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Indeed, if $g \neq 1$, left multiplication by g fixes no element of G .

Theorem 1.34. Let $W_1, \dots, W_r$ be all irreducible representations of G , and put $n_i = \dim W_i$. Then

$$V_{\text{reg}} \cong \bigoplus_{i=1}^r W_i^{\oplus n_i}.$$

Thus every irreducible representation occurs in the regular representation, with multiplicity equal to its degree.

Proof. Let χ_i be the character of W_i . By the multiplicity formula, its multiplicity in the regular representation is

$$\begin{aligned} \langle \chi_{\text{reg}}, \chi_i \rangle &= \frac{1}{|G|} \sum_{g \in G} \chi_{\text{reg}}(g) \overline{\chi_i(g)} \\ &= \chi_i(1) = n_i. \end{aligned}$$

□

Corollary 1.35. The degrees of the irreducible representations satisfy

$$\sum_{i=1}^r n_i^2 = |G|.$$

Moreover, for every $g \neq 1$,

$$\sum_{i=1}^r n_i \chi_i(g) = 0.$$

Proof. Taking dimensions in Theorem 1.34 gives the first identity. Taking characters gives

$$\chi_{\text{reg}} = \sum_{i=1}^r n_i \chi_i,$$

and evaluating at $g \neq 1$ gives the second. □

1.8 Class Functions and Character Tables

Definition 1.36. A function $f: G \rightarrow \mathbb{C}$ is called a *class function* if

$$f(hgh^{-1}) = f(g)$$

for all $g, h \in G$. Thus a class function is constant on every conjugacy class. Every character is a class function.

Let $\mathcal{H}$ denote the vector space of class functions on G .

Proposition 1.37. Let f be a class function and let V be a representation with character χ . Define

$$T_f = \sum_{g \in G} f(g) \rho(g).$$

Then T_f commutes with the action of G . If V is irreducible of degree n , then

$$T_f = \lambda \text{id}_V,$$

where

$$\lambda = \frac{1}{n} \sum_{g \in G} f(g) \chi(g) = \frac{|G|}{n} \langle f, \chi_{V^*} \rangle.$$

Proof. For $h \in G$,

$$\begin{aligned}\rho(h)^{-1}T_f\rho(h) &= \sum_{g \in G} f(g)\rho(h^{-1}gh) \\ &= \sum_{u \in G} f(huh^{-1})\rho(u) \\ &= T_f.\end{aligned}$$

Thus T_f is an intertwining operator. If V is irreducible, Schur's lemma gives $T_f = \lambda \text{id}_V$. Taking traces yields

$$\lambda n = \sum_{g \in G} f(g)\chi(g).$$

Since $\chi_{V^*}(g) = \overline{\chi(g)}$, we have

$$\langle f, \chi_{V^*} \rangle = \frac{1}{|G|} \sum_{g \in G} f(g)\chi(g),$$

which proves the formula. $\square$

Theorem 1.38. *The irreducible characters $\chi_1, \dots, \chi_r$ form an orthonormal basis of $\mathcal{H}$.*

Proof. They are orthonormal by Theorem 1.19. It remains to prove that they span $\mathcal{H}$.

Suppose that $f \in \mathcal{H}$ is orthogonal to every irreducible character. The dual of an irreducible representation is again irreducible, so f is also orthogonal to every dual character. By Proposition 1.37, the operator T_f acts as zero on every irreducible representation. Complete reducibility therefore implies that $T_f = 0$ on the regular representation.

Let e_1 be the basis vector of the regular representation corresponding to the identity. Then

$$0 = T_f e_1 = \sum_{g \in G} f(g)e_g.$$

Since the e_g form a basis, $f(g) = 0$ for every $g \in G$. Hence the orthogonal complement of the span of the irreducible characters is zero, so they form a basis. $\square$

Theorem 1.39. *The number of isomorphism classes of irreducible representations of G is equal to the number of conjugacy classes of G .*

Proof. A class function is determined uniquely by its value on each conjugacy class. Therefore the dimension of $\mathcal{H}$ is the number of conjugacy classes. By Theorem 1.38, the dimension of $\mathcal{H}$ is also the number of irreducible characters. $\square$

1.8.1 Column orthogonality

For $s \in G$, denote its conjugacy class by $\text{Cl}(s)$ and its centralizer by

$$C_G(s) = \{g \in G \mid gs = sg\}.$$

Recall that

$$|\text{Cl}(s)| = \frac{|G|}{|C_G(s)|}.$$

Theorem 1.40 (Column orthogonality). *Let $\chi_1, \dots, \chi_r$ be all irreducible characters. For $s, t \in G$,*

$$\sum_{i=1}^r \overline{\chi_i(s)} \chi_i(t) = \begin{cases} |C_G(s)|, & t \text{ is conjugate to } s, \\ 0, & t \text{ is not conjugate to } s. \end{cases}$$

In particular,

$$\sum_{i=1}^r |\chi_i(s)|^2 = |C_G(s)| = \frac{|G|}{|\text{Cl}(s)|}.$$

Proof. Let δ_s be the indicator function of the conjugacy class of s :

$$\delta_s(t) = \begin{cases} 1, & t \in \text{Cl}(s), \\ 0, & t \notin \text{Cl}(s). \end{cases}$$

Since the irreducible characters form an orthonormal basis of $\mathcal{H}$,

$$\delta_s = \sum_{i=1}^r \langle \delta_s, \chi_i \rangle \chi_i.$$

Now

$$\begin{aligned} \langle \delta_s, \chi_i \rangle &= \frac{1}{|G|} \sum_{g \in \text{Cl}(s)} \overline{\chi_i(g)} \\ &= \frac{|\text{Cl}(s)|}{|G|} \overline{\chi_i(s)} \\ &= \frac{1}{|C_G(s)|} \overline{\chi_i(s)}. \end{aligned}$$

Hence

$$|C_G(s)| \delta_s(t) = \sum_{i=1}^r \overline{\chi_i(s)} \chi_i(t),$$

which is the desired relation. □

Definition 1.41. Choose representatives $c_1, \dots, c_r$ of the conjugacy classes and list the irreducible characters $\chi_1, \dots, \chi_r$. The square matrix

$$(\chi_i(c_j))_{1 \leq i, j \leq r}$$

is called the *character table* of G .

The row orthogonality relations are

$$\sum_{j=1}^r |\text{Cl}(c_j)| \chi_i(c_j) \overline{\chi_k(c_j)} = |G| \delta_{ik},$$

while Theorem 1.40 gives the column orthogonality relations.

1.9 Induced Representations

Let $H \leq G$ and let

$$\theta: H \rightarrow \mathrm{GL}(W)$$

be a representation of H .

Definition 1.42. Suppose $\rho: G \rightarrow \mathrm{GL}(V)$ is a representation such that W is an H -invariant subspace of V and the restricted action on W is θ . For $g \in G$, set

$$W_{gH} = \rho(g)W.$$

This depends only on the left coset gH , because

$$\rho(gh)W = \rho(g)\rho(h)W = \rho(g)W$$

for $h \in H$. We say that V is *induced by* W , or that ρ is *induced by* θ , if

$$V = \bigoplus_{gH \in G/H} W_{gH}.$$

We write

$$V \cong \mathrm{Ind}_H^G W.$$

Equivalently, if R is a set of representatives of G/H , then

$$V = \bigoplus_{r \in R} \rho(r)W.$$

It follows immediately that

$$\dim \mathrm{Ind}_H^G W = [G : H] \dim W.$$

Example 1.43. The permutation representation of G on G/H is induced by the trivial representation of H :

$$\mathbb{C}[G/H] \cong \mathrm{Ind}_H^G \mathbf{1}_H.$$

Indeed, the one-dimensional subspace corresponding to the coset H is H -invariant and affords the trivial representation, and its translates are the basis lines corresponding to all cosets.

Example 1.44. The regular representation of G is induced by the regular representation of H . The subspace spanned by $(e_h)_{h \in H}$ is the regular representation of H , and its translates are the subspaces spanned by the basis vectors belonging to the left cosets of H .

Proposition 1.45. Suppose that V is induced by W , and let $W_1 \subseteq W$ be an H -invariant subspace. Then

$$V_1 = \bigoplus_{gH \in G/H} \rho(g)W_1$$

is a G -invariant subspace of V , and the representation on V_1 is induced by the representation of H on W_1 .

Proof. The sum is direct because it is contained term by term in the direct sum

$$V = \bigoplus_{gH \in G/H} \rho(g)W.$$

For $s \in G$, left multiplication sends the coset gH to sgH , and hence

$$\rho(s)\rho(g)W_1 = \rho(sg)W_1.$$

Thus $\rho(s)$ permutes the summands of V_1 , so V_1 is G -invariant. The defining direct-sum condition for induction is then automatic. $\square$

Lemma 1.46 (Universal property of induction). *Suppose that (V, ρ) is induced by (W, θ) , and let $\rho': G \rightarrow \text{GL}(V')$ be another representation. If $f: W \rightarrow V'$ is H -equivariant, then there exists a unique G -equivariant linear map*

$$F: V \rightarrow V'$$

whose restriction to W is f .

Proof. Choose a set R of representatives of G/H . Since

$$V = \bigoplus_{r \in R} \rho(r)W,$$

it is enough to define F on each summand. For $r \in R$ and $w \in W$, set

$$F(\rho(r)w) = \rho'(r)f(w).$$

This is well defined on the chosen direct-sum decomposition and clearly extends f when the representative of H is chosen to be 1.

We check G -equivariance. Let $g \in G$ and write

$$gr = r'h$$

with $r' \in R$ and $h \in H$. Then

$$\begin{aligned} F(\rho(g)\rho(r)w) &= F(\rho(r')\theta(h)w) \\ &= \rho'(r')f(\theta(h)w) \\ &= \rho'(r')\rho'(h)f(w) \\ &= \rho'(g)\rho'(r)f(w) \\ &= \rho'(g)F(\rho(r)w). \end{aligned}$$

Thus F is G -equivariant. Uniqueness follows because every element of V is a sum of vectors of the form $\rho(r)w$, and equivariance forces

$$F(\rho(r)w) = \rho'(r)F(w) = \rho'(r)f(w).$$

$\square$

Theorem 1.47 (Existence and uniqueness of induced representations). *For every representation $\theta: H \rightarrow \text{GL}(W)$, there exists a representation of G induced by θ . It is unique up to a unique G -equivariant isomorphism which restricts to the identity on W .*

Proof. Choose a set R of representatives of the left cosets G/H , with $1 \in R$, and let

$$V = \bigoplus_{r \in R} W_r,$$

where each W_r is a copy of W . For $g \in G$ and $r \in R$, write uniquely

$$gr = r'h$$

with $r' \in R$ and $h \in H$. Define

$$\rho(g)(w \in W_r) = \theta(h)w \in W_{r'}.$$

This is linear and invertible. To check the homomorphism property, suppose

$$kr = r_1h_1, \quad gr_1 = r_2h_2.$$

Then

$$gkr = r_2h_2h_1,$$

and therefore

$$\begin{aligned} \rho(g)\rho(k)(w \in W_r) &= \theta(h_2)\theta(h_1)w \in W_{r_2} \\ &= \theta(h_2h_1)w \in W_{r_2} \\ &= \rho(gk)(w \in W_r). \end{aligned}$$

Thus ρ is a representation. The summand W_1 is H -invariant and the action of H on it is θ , while its G -translates are precisely the summands W_r . Hence V is induced by W .

For uniqueness, let V and V' be two representations induced by the same H -representation W . Applying Lemma 1.46 to the inclusion $W \hookrightarrow V'$ gives a G -equivariant map $F: V \rightarrow V'$ extending the identity on W . Similarly, there is a G -equivariant map $F': V' \rightarrow V$ extending the identity on W . Both $F'F$ and FF' extend the identity on W ; by the uniqueness part of the universal property, they are the identity maps. Thus F is the required isomorphism, and its uniqueness follows from the same universal property. $\square$

Theorem 1.48 (Character of an induced representation). *Let R be a set of representatives of G/H , and let χ_θ be the character of W . If χ_{Ind} is the character of $\text{Ind}_H^G W$, then for every $u \in G$,*

$$\chi_{\text{Ind}}(u) = \sum_{\substack{r \in R \\ r^{-1}ur \in H}} \chi_\theta(r^{-1}ur).$$

Equivalently,

$$\chi_{\text{Ind}}(u) = \frac{1}{|H|} \sum_{\substack{s \in G \\ s^{-1}us \in H}} \chi_\theta(s^{-1}us).$$

Proof. With respect to the decomposition

$$\text{Ind}_H^G W = \bigoplus_{r \in R} W_r,$$

the operator $\rho(u)$ sends W_r to the summand corresponding to the coset urH . A summand contributes to the trace only when it is fixed, that is, when

$$urH = rH,$$

or equivalently $r^{-1}ur \in H$. On such a summand, the action of u is, after identifying W_r with W , the operator $\theta(r^{-1}ur)$. Therefore its contribution to the trace is $\chi_\theta(r^{-1}ur)$, proving the first formula.

For the second formula, every left coset rH has $|H|$ elements. If $s = rh$, then

$$s^{-1}us = h^{-1}(r^{-1}ur)h,$$

and χ_θ is constant on conjugacy classes in H . Hence all $|H|$ elements of the coset give the same contribution, so summing over all $s \in G$ multiplies the first sum by $|H|$. $\square$